

Hexie (train)

Hexie (Harmony)



CRH2B at Songjiang South railway station

<u>In service</u>	<u>28 January 2007 – Present</u>
<u>Manufacturers</u>	<u>Bombardier Transportation</u> <u>Kawasaki Heavy Industries</u> <u>Alstom</u> <u>Siemens</u> <u>CRRC</u>
<u>Operator</u>	<u>China Railway Corporation</u>

Specifications

<u>Train length</u>	<u>200.84 m (658 ft 11 in) or more</u>
<u>Maximum speed</u>	<u>380 km/h (236 mph)</u>
<u>Power supply</u>	<u>Overhead catenary</u>
<u>Electric system</u>	<u>25 kV 50 Hz AC</u>
<u>Current collection</u>	<u>Pantograph</u>
<u>Track gauge</u>	<u>1,435 mm (4 ft 8½ in) standard gauge</u>

Hexie

<u>Simplified Chinese</u>	和谐号
<u>Traditional Chinese</u>	和諧號

Hexie (Chinese: 和谐号; pinyin: *Héxié hào*; lit. 'Harmony'), also known as the **CRH** series EMU, is an umbrella term for the multiple unit high-speed and higher-speed trains operated by China Railway under the China Railway High-speed brand. All series of Hexie are based on foreign-developed technology and later manufactured locally in China through technology transfer licenses, with the ultimate goal of China acquiring the know-how and capability to produce high-speed rail trains.

The Harmony series does not belong to any single platform, instead encompassing all high-speed trains in China with roots in foreign technology, specifically CRH1, CRH2, CRH3 and CRH5. Although later variants of Hexie such as CRH380A were designed by Chinese companies, they are still classified as CRH due to incorporation of foreign technology.

History

In 2007, China's Ministry of Railways drafted a plan for China's future high-speed network. Bombardier Transportation, Kawasaki Heavy Industries, Alstom and later Siemens joined the high-speed train manufacturing project that later became known as Harmony. Forming joint-ventures with Chinese company CNR and CSR, these four foreign companies signed agreements with China to manufacture high-speed trains for China as well as provide assistance for Chinese companies to manufacture train cars locally in the future.

While the initial train sets from each Hexie series were manufactured overseas, subsequent sets are manufactured locally through technology transfer, a key requirement for China. The signalling, track and support structures, control software, and station design are developed domestically with foreign elements as well. Although the first domestically produced trains were initially delivered in complete knock-down form, eventually manufacturing as a whole became predominantly Chinese.^[1]

China currently holds many patents related to the internal components of these trains, re-designed in China to allow the trains to run at higher speeds than the foreign designs allowed. However, these patents are only valid within China, and as such hold no international power. This weakness on the intellectual property of Hexie trains became an obstruction for China to export its high-speed rail related products,

leading to the development of a completely redesigned train brand, Fuxing, which is based on indigenous technologies.^{[1][2][3]}

While most Hexie series trains are designed for China Railway, a variant of the CRH380A has been modified for operation by MTR Corporation of Hong Kong, who will operate these trains under the brand Vibrant Express specifically for the Guangzhou–Shenzhen–Hong Kong Express Rail Link.

Variants



China Railway High-speed train passing through Shenzhou railway station in Hainan



Engineer's compartment on the CRH3

The *Harmony* brand has different electric multiple unit trainsets, the designs for which are imported from other nations and designated CRH-1 through CRH-5 and CRH380A(L), CRH380B(L), and CRH380C(L). CRH trainsets are intended to provide fast and convenient travel between cities. Some of the trainsets are manufactured locally through technology transfer, a key requirement for China. The signalling, track and support structures, control software, and station design are developed domestically with foreign elements as well, so the system as a whole is predominantly Chinese.^[1] China currently holds many new patents related to the

internal components of these trains, re-designed in China to allow the trains to run at higher speeds than the foreign designs allowed. However, these patents are only valid within China, and as such hold no international power.^{[1][2][3]}

The trainsets are as follows:

- CRH1 produced by Bombardier Transportation's joint venture Sifang Power (Qingdao) Transportation (BST), CRH1A, and CRH1B, nicknamed "Metro" or "Bread", derived from Bombardier's Regina; CRH1E, nicknamed "Lizard", is Bombardier's ZEFIRO 250 design.
 - CRH1A: sets consists of 8 cars; maximum operating speed of 250 km/h (155 mph)
 - CRH1B: a modified 16-car version; maximum operating speed of 250 km/h (155 mph)
 - CRH1E: a 16-car high-speed sleeper version; maximum operating speed of 250 km/h (155 mph)
- CRH2: nicknamed "Hairtail", derived from E2 Series 1000 Shinkansen.
 - CRH2A: In 2006, China unveiled CRH2, a modified version of the Japanese Shinkansen E2-1000 series. An order for 60 8-car sets had been placed in 2004, with the first few built in Japan, the rest produced by Sifang Locomotive and Rolling Stock in China.^[4]
 - CRH2B: a modified 16-car version of CRH2; maximum operating speed of 250 km/h (155 mph)
 - CRH2C (Stage one): a modified version of CRH2 with a maximum operating speed up to 300 km/h (186 mph) as a result of replacing two intermediate trailer cars with motored cars
 - CRH2C (Stage two): a modified version of CRH2C (stage one) has a maximum operating speed up to 350 km/h (217 mph) by using more powerful motors
 - CRH2E: a modified 16-car version of CRH2 with sleeping cars
- CRH3: nickname "Rabbit", derived from Siemens ICE3 (class 403); 8-car sets; maximum operating speed of 350 km/h (217 mph).
- CRH5A: derived from Alstom Pendolino ETR600; 8-car sets; maximum operating speed of 250 km/h (155 mph).^[5]
- CRH6: designed by CSR Puzhen and CSR Sifang, will be manufactured

by CSR Jiangmen. It is designed to have two versions: one with a top operating speed of 220 km/h (137 mph); the other with a top operating speed of 160 km/h (99 mph). They will be used on 200 km/h (124 mph) or 250 km/h (155 mph) Inter-city High Speed Rail lines; planned to enter service by 2011.

- CRH380A; Maximum operating speed of 380 km/h (236 mph). Developed by CSR and manufactured by Sifang Locomotive and Rolling Stock; entered service in 2010.
 - CRH380A: 8-car version
 - CRH380AL: 16-car version
- CRH380B: upgraded version of CRH3; maximum operating speed of 380 km/h (236 mph), manufactured by Tangshan Railway Vehicle and CRRC Changchun Railway Vehicles; entered service in 2011.
 - CRH380B: 8-car version
 - CRH380BL: 16-car version
- CRH380CL: designed and manufactured by CRRC Changchun Railway Vehicles. Maximum operating speed of 380 km/h; planned to enter service in 2012.
- CRH380D: also named Zefiro 380; maximum operating speed of 380 km/h (236 mph), manufactured by Bombardier Sifang (Qingdao) Transportation Ltd.; planned to enter service in 2012.
 - CRH380D: 8-car version

CRH1A, B,E, CRH2A, B,E, and CRH5A are designed for a maximum operating speed (MOR) of 200 km/h (124 mph) and can reach up to 250 km/h (155 mph). CRH3C and CRH2C designs have an MOR of 310 km/h (193 mph), and can reach up to 350 km/h (217 mph), with a top testing speed more than 380 km/h (236 mph). However, in practical terms, issues such as maintenance costs, comfort, and safety make the maximum speed of more than 380 km/h (236 mph) impractical and remain limiting factors.

Train type	Car dimensions	Total length	Top speed	Seating capacity
CRH1				
<u>CRH1A-200</u>	End cars length:26.95 m (88 ft 5 in) Inter cars length:26.6 m (87 ft 3 in) Width:3.328 m (10 ft 11.0 in) Height:4,040 mm (13 ft 3 in)	Calculated:213.5 m (700 ft 6 in) Real:213.5 m (700 ft 6 in)	Test: 278 km/h (173 mph) Design: 255 km/h (158 mph) Continuous operation:200 km/h (124 mph) Current operation:210 km/h (130 mph)	673: 144 first and 529 standard 612: 128 first and 484 standard
<u>CRH1A-250</u>			Test: 278 km/h (173 mph) Design: 285 km/h (177 mph) Continuous operation:250 km/h (155 mph) Current operation:250 km/h (155 mph)	597: 144 first and 453 standard 646: 128 first and 518 standard 635: 128 first and 507 standard 673: 144 first and 529 standard
<u>CRH1B</u>		Calculated:426.3 m (1,398 ft 7 in) Real:426.3 m (1,398 ft 7 in)	Test: 292 km/h (181 mph) Design: 275 km/h (171 mph) Continuous operation:250 km/h (155 mph) Current operation:250 km/h (155 mph)	1299: 208 first and 1091 standard
<u>CRH1B (1E head)</u>	End cars length:28.28 m (92 ft 9 in) Inter cars length:26.6 m	Calculated:428.96 m (1,407 ft 4 in) Real:428.9 m (1,407 ft 2 in)		1299: 208 first and 1091 standard

<u>CRH1E</u>	(87 ft 3 in) Width:3.328 m (10 ft 11.0 in) Height:4.04 m (13 ft 3 in)		Test: N/A Design: 275 km/h (171 mph) Continuous operation:250 km/ h (160 mph) Current operation:250 km/ h (160 mph)	618: 122 standard, 16 luxury sleepers and 480 soft sleepers 642: 122 standard and 520 soft sleepers
<u>CRH1A-A</u>	End cars length:26.995 m (88 ft 6.8 in) Inter cars length:25.325 m (83 ft 1.0 in) Width:3.358 m (11 ft 0.2 in) Height:4.16 m (13 ft 8 in)	Calculated:205.94 m (675 ft 8 in) Real:207.9 m (682 ft 1 in)		613: 48 fir and 565 standard
<u>CRH1E- 250</u>		Calculated:408.54 m (1,340 ft 4 in) Real:413.1 m (1,355 ft 4 in)		642: 110 standard and 532 soft sleepers
<u>CRH380D</u>	End cars length:27.85 m (91 ft 4 in) Inter cars length:26.6 m (87 ft 3 in) Width:3.368 m (11 ft 0.6 in) Height:4.16 m (13 ft 8 in)	Calculated:215.3 m (706 ft 4 in) Real:215.3 m (706 ft 4 in)	Test: 420 km/h (260 mph) Design: 422 km/h (262 mph) Continuous operation:380 km/ h (240 mph) Current operation:350 km/ h (220 mph)	554: 10 sightseein 32 first an 512 standard 556: 10 business, 28 first an 518 standard
CRH2				
<u>CRH2G</u>	End cars length:25.7 m (84 ft 4 in) Inter cars length:25 m (82 ft 0 in) Width:3.33 m (10 ft 11 in) Height:3.86 m	Calculated:201.4 m (660 ft 9 in) Real:201.4 m (660 ft 9 in)	Test: N/A Design: 275 km/h (171 mph) Continuous operation:250 km/ h (160 mph) Current operation:250 km/ h (160 mph)	613: 48 fir and 565 standard

	(12 ft 8 in)			
<u>CRH2A</u> <u>(EC)</u>	End cars length:25.7 m (84 ft 4 in) Inter cars length:25 m (82 ft 0 in) Width: 3.38 m (11 ft 1 in) Height:3.7 m (12 ft 2 in)		Test: N/A Design: 300 km/h (190 mph) Continuous operation:250 km/ h (155 mph) Current operation:250 km/ h (155 mph)	
<u>CRH2A</u> <u>(Old)</u>			Test: 282 km/h (175 mph) Design: 300 km/h (190 mph) Continuous operation:250 km/ h (160 mph) Current operation:250 km/ h (160 mph)	
<u>CRH2C</u> <u>Stage 1</u>			Test: 394.2 km/h (244.9 mph) Design: 370 km/h (230 mph) Continuous operation:320 km/ h (200 mph) Current operation:300 km/ h (190 mph)	610: 51 fir and 559 standard
<u>CRH2C</u> <u>Stage 2</u>			Test: N/A Design:394 km/h (245 mph) Continuous operation:350 km/ h (217 mph) Current operation:310 km/ h (193 mph)	
<u>CRH2B</u>		Calculated:401.4 m (1,316 ft 11 in) Real:401.4 m	Test: 275 km/h (171 mph) Design:300 km/h	1240: 155 first and 1075

			(190 mph) Continuous operation:250 km/ h (155 mph) Current operation:250 km/ h (160 mph)	standard
<u>CRH2E</u>		(1,316 ft 11 in)	Test: N/A Design:275 km/h (171 mph) Continuous operation:250 km/ h (155 mph) Current operation:250 km/ h (155 mph)	630: 110 standard and 520 soft sleepers
<u>CRH2E- NG</u>	End cars length:25.7 m (84 ft 4 in) Inter cars length:25 m (82 ft 0 in) Width:3.33 m (10 ft 11 in) Height:3.75 m (12 ft 4 in)			642: 110 standard and 532 soft sleepers
	End cars length:26.825 m (88 ft 0.1 in) Inter cars length:25.65 m (84 ft 2 in) Width:3.33 m (10 ft 11 in) Height:4,050 mm (13 ft 3 in)	Calculated:412.75 m (1,354 ft 2 in) Real:412.8 m (1,354 ft 4 in)	Test: N/A Design:280 km/h (174 mph) Continuous operation:250 km/ h (160 mph) Current operation:250 km/ h (160 mph)	880: 880 soft sleepers
<u>CRH380A</u>	End cars length:26.5 m (86 ft 11 in) Inter cars length:25 m (82 ft 0 in) Width:3.38 m (11 ft 1 in) Height:3.7 m (12 ft 2 in)	Calculated:203 m (666 ft 0 in) Real:203 m (666 ft 0 in)	Test:416.6 km/h (258.9 mph) Design:416.6 km/ h (258.9 mph) Continuous operation:380 km/ h (240 mph) Current operation:350 km/ h (220 mph)	480: 18 sightseein 89 first an 373 standard

<u>CRH380A</u> (New)			Test: N/A Design: 420 km/h (260 mph) Continuous operation:380 km/h (240 mph) Current operation:350 km/h (220 mph)	556: 10 business, 28 first an 518 standard
<u>CRH380AL</u>		Calculated:403 m (1,322 ft 2 in) Real:403 m (1,322 ft 2 in)	Test: 486.1 km/h (302.0 mph) Design: 420 km/h (260 mph) Continuous operation:380 km/h (240 mph) Current operation:350 km/h (220 mph)	1028: 28 business, 162 first and 838 standard 1061: 20 business, 118 first and 923 standard

CRH3

<u>CRH3A</u>	End cars length:27.05 m (88 ft 9 in) Inter cars length:25 m (82 ft 0 in) Width:3.3 m (10 ft 10 in) Height: 3.9 m (12 ft 10 in)	Calculated:204.1 m (669 ft 7 in) Real:209.75 m (688 ft 2 in)	Test: N/A Design: 420 km/h (260 mph) Continuous operation:380 km/h (240 mph) Current operation:310 km/h (190 mph)	613: 48 fir and 565 standard
<u>CRH3C</u>	End cars length:25.86 m (84 ft 10 in) Inter cars length:24.825 m (81 ft 5.4 in) Width:3.265 m (10 ft 8.5 in) Height:3,890 mm (12 ft 9 in)	Calculated:200.67 m (658 ft 4 in) Real:200.67 m (658 ft 4 in)	Test: 394.3 km/h (245.0 mph) Design:404 km/h (251 mph) Continuous operation:350 km/h (217 mph) Current operation:310 km/h (193 mph)	546: 16 special, 40 first and 490 standard

CRH380BG	End cars length:25.86 m (84 ft 10 in) Inter cars length:24.825 m (81 ft 5.4 in)		Test: N/A Design: 420 km/h (260 mph) Continuous operation:380 km/h (236 mph) Current operation:350 km/h (217 mph)	551: 16 sightseein 44 first an 491 standard 556: 10 sightseein 28 first an 518 standard
CRH380BL	Width:3.257 m (10 ft 8.2 in) Height:3,890 mm (12 ft 9 in)	Calculated:399.27 m (1,309 ft 11 in) Real:399.27 m (1,309 ft 11 in)	Test: 487.3 km/h (302.8 mph) Design: 487 km/h (303 mph) Continuous operation:380 km/h (240 mph) Current operation:350 km/h (217 mph)	1005: 28 business, 186 first and 791 standard 1015: 26 business, 118 first and 871 standard
CRH380B	End cars length:26.475 m (86 ft 10.3 in) Inter cars length:25 m (82 ft 0 in) Width:3.257 m (10 ft 8.2 in) Height:3,890 mm (12 ft 9 in)	Calculated:202.95 m (665 ft 10 in) Real:202.95 m (665 ft 10 in)	Test: N/A Design: 446 km/h (277 mph) Continuous operation:380 km/h (240 mph) Current operation:350 km/h (220 mph)	556: 10 business, 28 first an 518 standard
CRH380CL	End cars length:26.525 m (87 ft 0.3 in) Inter cars length:24.825 m (81 ft 5.4 in) Width:3.257 m (10 ft 8.2 in) Height:3,890 mm (12 ft 9 in)	Calculated:400.6 m (1,314 ft 4 in) Real:400.27 m (1,313 ft 3 in)	Test: N/A Design:420 km/h (261 mph) Continuous operation:380 km/h (236 mph) Current operation:350 km/h (217 mph)	1029: 28 business, 130 first and 871 standard 1015: 26 business, 118 first and 871 standard
CRH5				

<u>CRH5A</u>	End cars length:27.6 m (90 ft 7 in) Inter cars length:25 m (82 ft 0 in) Width:3.2 m (10 ft 6 in) Height:4.27 m (14 ft 0 in)	Calculated:205.2 m (673 ft 3 in) Real:211.5 m (693 ft 11 in)	Test: N/A Design: 275 km/h (171 mph) Continuous operation:250 km/ h (160 mph) Current operation:250 km/ h (160 mph)	622: 60 fir and 562 standard 586: 112 first and 474 standard
<u>CRH5G (Old)</u>			Test: N/A Design:275 km/h (171 mph) / 300 km/h (186 mph) Continuous operation:250 km/ h (160 mph) Current operation:250 km/ h (160 mph)	613: 48 fir and 565 second class
<u>CRH5G (New)</u>	End cars length:26.85 m (88 ft 1 in) Inter cars length:26 m (85 ft 4 in) Width:3.3 m (10 ft 10 in) Height:4.1 m (13 ft 5 in)	Calculated:209.7 m (688 ft 0 in) Real:211.5 m (693 ft 11 in)		
<u>CRH5E</u>		Calculated:417.7 m (1,370 ft 5 in) Real:418.7 m (1,373 ft 8 in)	Test: N/A Design: 275 km/h (171 mph) Continuous operation:250 km/ h (160 mph) Current operation:250 km/ h (160 mph)	642: 110 standard and 532 soft sleepers 880: 880 soft sleepers
CRH6				
<u>CRH6A</u>	End cars length:25.7 m (84 ft 4 in) Inter cars length:25 m (82 ft 0 in) Width:3.3 m (10 ft 10 in) Height:3,860 mm (12 ft 8 in)	Calculated:201.4 m (660 ft 9 in) Real:201.4 m (660 ft 9 in)	Test: 220 km/h (140 mph) Design: 250 km/h (160 mph) Continuous operation:200 km/ h (120 mph) Current operation:200 km/ h (120 mph)	1488: 557 standard and 931 standings 1488: 477 standard and 1011 standings 613: 48 fir and 565 standard

<u>CRH6A-A</u>		Calculated:101.4 m (332 ft 8 in) Real:101.4 m (332 ft 8 in)		688: 252 standard and 436 standings
<u>CRH6F</u>		Calculated:201.4 m (660 ft 9 in) Real:201.4 m (660 ft 9 in)	Test: 176 km/h (109 mph) Design: 200 km/h (124 mph) Continuous operation:160 km/ h (99 mph) Current operation:160 km/ h (99 mph)	1988: ... standard and ... standings
<u>CRH6F-A</u>		Calculated:101.4 m (332 ft 8 in) Real:101.4 m (332 ft 8 in)		875: ... standard and ... standings

Chinese CRH trainsets order timetable

Date	Factory	Speed Level	Type	Quantity (set)	Quantity (car)	
2004-10-10 ^[6]	Alstom	250 km/h	CRH5A	3	24	6
	CNR Changchun			57	456	
2004-10-12 ^[7]	BST (Bombardier & CSR)	250 km/h	CRH1A	20	160	US
2004-10-20 ^[8]	Kawasaki	250 km/h	CRH2A	3	24	9
	CSR Sifang			57	456	
2005-05-30 ^[9]	BST	250 km/h	CRH1A	20	160	US
2005-06 ^[8]	CSR Sifang	300 km/h	CRH2C Stage one	30	240	8
		350 km/h	CRH2C Stage two	30	240	
2005-11-20 ^[10]	Siemens	350 km/h	CRH3C	3	24	13
	CNR Tangshan			57	456	
2007-10-31 ^[11]	BST	250 km/h	CRH1B	20	320	1
			CRH1E	20	320	
2007-11 ^[12]	CSR Sifang	250 km/h	CRH2B	10	160	1
2007-11 ^[8]	CSR Sifang	250 km/h	CRH2E	6	96	9
2008-12-06 ^[8]	CSR Sifang	250 km/h	CRH2E	14	224	2
2009-09-23 ^[13]	CNR Changchun	250 km/h	CRH5A	30	240	4
2009-03-16 ^[14]	CNR Tangshan	380 km/h	CRH380BL	70	1,120	39
	CNR Changchun			30	480	

Date	Factory	Speed Level	Type	Quantity (set)	Quantity (car)	
2009-09-28 ^[15]	CSR Sifang	380 km/h	CRH380A	40	320	45
			CRH380AL	100	1,600	
2009-09-28 ^[15] Modified 2012-09-05 ^[16]	BST	380 km/h	CRH380D	70	560	27
		250 km/h	CRH1A	46	368	
		250 km/h	Zefiro 250NG	60	480	
2009-09-28 ^[17]	CNR Changchun	380 km/h	CRH380B	40	320	23
			CRH380BL	15	240	
			CRH380CL	25	400	
2009-09-28 ^[18]	CNR Tangshan	350 km/h	CRH3C ^[19]	20	160	3,
2009-12-30 ^[20]	CSR Puzhen	220 km/h	CRH6	24	192	2,
2010-07-16 ^[21]	BST	250 km/h	CRH1A	40	320	5,
2010-09-14 ^[22]	CSR Sifang	250 km/h	CRH2A	40	320	3,
2010-10-13 ^[23]	CNR Changchun	250 km/h	CRH5A	20	160	2,
2011-04-26	CNR Changchun	250 km/h	CRH5A	30	240	3,
Total				1050	10,240	

Chinese CRH trainsets delivery timetable

Based on data published by Sinolink Securities;^{[24][25]} some small changes were made according to the most recent news.

Type	2006	2007	2008	2009	2010	2011	Future (plan)	Total
CRH1A	8	18	12	2	20	20	106	80
CRH2A	19	41			15	25		100
CRH5A		27	29	4	30	20	30	140
CRH1B			4	9	7			20
CRH1E				3	8	9		20
CRH2B			10					10
CRH2E			6	14				20
CRH2C			10	20	30			60
CRH3C			7	36	37			80
CRH380A					40			40
CRH380AL					6	94		100
CRH380B						20	20 ¹	40
CRH380BL					11	49	55 ¹	115
CRH380CL							25 ¹	25
CRH380D							70 ²	70
CRH380DL							0 ²	0
CRH6							24	24
Total	27	86	78	88	204	237	330	1000
Cumulative	27	113	191	279	483	744	1050	1000

- ¹ All CRH380B and CRH380C units to be delivered before 2012.
- ² All CRH380D units to be delivered before 2014.

Gallery



CRH1



Two coupled 8-car
CRH1A electric multiple
unit train sets in
Nanchang



CRH2



CRH2 First Class Coach



CRH2 Second Class
Coach



CRH2C at Shanghai
Hongqiao railway station



CRH3



Premier-class cabin
inside Beijing-Tianjin
Intercity CRH3 train



CRH2C and CRH3C at
Tianjin railway station



CRH5



CRH380A



CRH380BL Business
Coach



CRH380CL at Beijing
South railway station



CRH380D at Beijing
South railway station

See also

- China Railway High-speed, Chinese high-speed railway service provided by China Railway.
- China Railway, Chinese state-owned corporation that operates nearly all Harmony trains.
- Fuxing (train), next-generation EMU developed by China with independent intellectual property rights.

References

1. Shirouzu, Norihiko (2010-11-17). "Train Makers Rail Against China's High-Speed Designs" (<https://www.wsj.com/articles/SB10001424052748704814204575507353221141616>). *The Wall Street Journal*. Retrieved 2012-12-26.
2. Wines, Michael; Bradsher, Keith (2011-02-17). "China Rail Chief's Firing Hints at Trouble" (<https://www.nytimes.com/2011/02/18/world/asia/18rail.html?pagewanted=all&r=0>). *The New York Times*. Retrieved 2012-12-27. "Many multinational companies also resent China for tweaking foreign designs and building the equipment itself rather than importing it."